

Peter A. Lilley

Software Engineer · C++ · C#/.NET · Python · CAD/BIM and 3D graphics

Chesterfield / St. Louis, MO | pal@cadpal.net | Available on request | github.com/freeParameterized

Software engineer with a CAD/BIM background. Builds native desktop software in C++ (Qt 6, OpenGL, Open CASCADE), production automation in C#/.NET and Python, and web and mobile apps in TypeScript and Flutter. Two years at David Mason & Associates building Civil 3D tooling that engineers use daily. Built Cadnat, a CAD/BIM desktop application of about 80,000 lines of C++, and released Omabiblia, an open-source OpenGL application. Uses AI-assisted development alongside testing, CI, packaging, and code review.

EXPERIENCE

Automation Tooling Engineer (Staff Technician)

David Mason & Associates — St. Louis, MO

Jul 2024 - Aug 2026

- Engineered automated quality checks that cut an estimated 25% of drafting errors prior to engineer review.
- Built a generator for standard details and grading from existing survey data. Hundreds of thousands of entities; days of hand modeling became minutes.
- Wrote Dynamo automation that reads Civil 3D Excel reports and draws pipe flow levels and HGLs; about 8 hours of clicking fell to minutes.
- Built C#/.NET Civil 3D plugins, OCR ingestion of PDF plan sets, and an OpenAI pipeline, with Git pull-request review and CI/CD. Colleagues use the tooling daily.

Quality Engineer - Manufacturing / Precision Machining

Component Bar Products — O'Fallon, MO

Apr 2023 - Jun 2024

- Programmed CMMs and validated machined geometry to GD&T specs under ISO 9001 and PPAP; managed calibration and traceability.

Assistant Department Manager - Building Materials and Millwork

Menard, Inc. — Manchester / Ballwin, MO

Jun 2020 - Mar 2023

- Supervised department staff, priced large commercial orders, and turned contractor requests into material lists on deadline.

CAD Drafter

Jeff Day & Associates, LLC — Manchester, MO

2020

- Drafted architectural construction documents in AutoCAD; kept layers and the detail library to office CAD standards.

Revit/CAD Technician - Mechanical/Plumbing

Heideman & Associates, Inc. — Fenton, MO

May 2018 - Apr 2019

- Drafted mechanical and plumbing documents in Revit and AutoCAD (St. Luke's, Mercy, AT&T, SLU, American Water); clash detection across disciplines.

SELECTED PROJECTS

Cadnat — CAD/BIM desktop application

C++17 · Qt 6 · Open CASCADE · CMake/CTest · Python

- Native 2D/3D CAD and BIM app: about 80k lines of C++ on Qt 6 and Open CASCADE. DXF/DWG, STEP, IGES, glTF, and PDF I/O; local NDJSON automation API with 100+ undoable commands.
- Headless sweep of about 3,000 command and menu runs found 130 failures (9 crashes, 84 hangs), all fixed. About 20 CTest suites; GitHub Actions builds.
- Packaged as AppImage (tested on 10 distros), Flatpak, Windows, and macOS, with offline Ed25519-signed licensing.

Cadnat Generators — parametric drawing sets

Python · ezdxf · IFC · YAML / JSON Schema

- About 35k lines of Python turning YAML parameters into permit-style DXF sheet sets (houses in 43 styles, kitchens, baths, decks, stairs, trusses), with IFC2X3/IFC4/IFC4X3 export including terrain, alignments, and georeferencing.

Cadnat XR — LAN protocol and headset clients

C++ (Unreal Engine 5) · C# (Unity 6) · Python · TCP/UDP

- Designed CADNET/1 (UDP discovery, NDJSON over TCP, HTTP assets): a Python hub serving CAD models to Meta Quest 3 clients in Unreal Engine 5 and Unity 6, with live pose relay and 63 automated tests.

Omabiblia — open-source OpenGL application

C++20 · OpenGL 3.3 / GLSL · Dear ImGui · GLFW · Wayland

- Offline desktop app: five real-time OpenGL scenes, CRT post-processing (bloom, ACES), a chiptune synthesizer, a shell with pipes, and a click-through Wayland pop-out. CTest suites and prebuilt releases.

Digital Twin Pro

Flutter · Dart 3 · SQLite · Firebase · ARCore

- Flutter 3D inventory app with a hand-written renderer (Z-sorted draw queue, custom projection); voice, photo, and typed input feed one typed data model. Google Play beta.

OraLab

TypeScript · Next.js · Java / H2 Oracle mode · Monaco

- Shipped a public Oracle SQL training lab: live engine, classic HR sample, Monaco editor, tutorials, and a docked schema graph.

CAD integration bridge

C++17 · Dear ImGui · Windows COM

- C++17 host that drives BricsCAD, AutoCAD, and Civil 3D over COM, finds installed SDKs at runtime, and hot-loads LISP, C#, and native plugins.

SKILLS

Languages C++ (17/20), C# / .NET, Python, TypeScript, JavaScript, GLSL, Dart, SQL, Oracle SQL, LISP, VBA

Desktop, graphics, and geometry Qt 6 (Widgets, RHI), OpenGL 3.3 / GLSL, Dear ImGui, GLFW, Open CASCADE (B-rep, STEP, IGES), DXF/DWG, IFC2X3/IFC4/IFC4X3, glTF, Wayland, Unreal Engine 5 and Unity 6 (Meta Quest 3)

Software engineering Object-oriented design, REST and NDJSON APIs, TCP/UDP protocol design, multi-threading, CMake/CTest, GitHub Actions CI/CD, Git (GitFlow, trunk-based, pull requests), AppImage/Flatpak packaging, Linux, Docker, Bash

Databases and data Postgres, SQLite, Oracle SQL, Firebase, H2, Cloudflare D1

CAD and domain AutoCAD, Civil 3D, Revit, Navisworks, Dynamo, CAD plugins, typed metadata, GD&T, CMM, ISO 9001

EDUCATION

Ranken Technical College

A.A.S. in Building Systems Engineering Technology

2019 · Wentzville, MO

Additional coursework: Technical Spanish and Applications (Missouri S&T), Oracle Database SQL (Oracle University).